

MINOR PROJECT SYNOPSIS(PR1103)

on Academic Performance Evaluator & Learning Management LMS

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1. INTRODUCTION

In many educational institutions, academic performance evaluation and monitoring are still carried out manually or using basic tools such as spreadsheets. Although these methods store student data, they fail to provide continuous monitoring, meaningful analysis, or timely feedback. This creates difficulties for teachers in tracking student progress and for students in understanding their academic strengths and weaknesses. The problem becomes more severe in large classrooms, where individual performance tracking is time-consuming and error-prone.

To overcome these limitations, this project proposes an Academic Performance Evaluator and Learning Management System (LMS). The system provides a centralized digital platform where students can access grades, track performance and obtain learning resources, while teachers can efficiently evaluate, monitor and guide students. By automating evaluation and integrating learning management features, the system reduces manual workload, improves transparency and enhances the overall teaching–learning process.

2. MOTIVATION

The primary motivation for this project is the inefficiency of traditional academic evaluation systems. Manual evaluation methods are slow, prone to errors and unsuitable for large student groups. Existing LMS platforms mainly focus on content delivery and lack advanced performance analysis, personalized feedback and integrity monitoring during online assessments.

Students often receive delayed feedback, struggle to manage multiple academic activities and feel demotivated due to unclear performance insights. Teachers, on the other hand, face heavy administrative workloads and difficulty in monitoring individual progress. Online quizzes and assessments are also vulnerable to unfair practices, further reducing trust in digital evaluation systems.

This project aims to address these issues by developing an LMS that automates performance tracking, supports timely feedback, improves communication and ensures efficient data management. The system seeks to enhance academic efficiency, student engagement and reliability of evaluation in a unified platform.

3. LITERATURE REVIEW

3.1 Adaptive LMS and Student Engagement: A Correlational Study

Tamayo (2025) studied the impact of an adaptive LMS on student academic performance using a rule-based system developed through the ADDIE model. The study revealed that time spent on the LMS and number of pages viewed had a strong positive correlation with academic performance, while login frequency showed minimal impact. The research emphasized the importance of structured content delivery, evaluation modules and interactive LMS features to enhance student engagement. However, it also highlighted challenges such as sustaining engagement and offering personalized learning paths in large classrooms.

Naïve Approach:

Our project initially focuses on a basic LMS that stores academic records, provides access to learning materials and tracks essential student activity. This creates a structured foundation that can later be enhanced with adaptive and analytical features.

3.2 Advanced Educational Data Mining in LMS

Zhang et al. (2024) proposed an LMS integrated with an educational data mining module to extract meaningful insights from student performance data. The study introduced the Faster Apriori algorithm, which reduces computational complexity by dynamically pruning irrelevant itemsets. The research demonstrated that mining both common and exception rules helps identify critical academic performance factors, such as study habits and prior academic history, enabling targeted educational interventions.

Naïve Approach:

Inspired by this work, our system begins with simple automated performance tracking, which can later be extended to support data-driven academic insights and targeted interventions.

3.3 Predictive Analytics for At-Risk Students

Guo et al. (2022) applied machine learning techniques on LMS data to predict academically at-risk students. Using data from the Brightspace LMS, the study found that Random Forest achieved the highest accuracy (94.75%). Feature importance analysis using SHAP values identified assignment submissions and content completion as the strongest predictors of academic success, while total time spent on content had minimal influence. The study highlights the importance of proactive intervention rather than passive grade recording.

Naïve Approach:

Based on these findings, our project prioritizes automated tracking of key academic indicators such as assessments and course completion to support early identification of underperforming students.

4. PROBLEM STATEMENT

- Many colleges still check student performance by hand or using Excel sheets
- This way is slow, confusing and has many mistakes, especially when there are many students
- Teachers get extra work and cannot give feedback to students on time
- Students cannot clearly see their marks, progress, or weak subjects
- Current LMS systems mostly only share study material and do not properly check performance automatically
- These systems also do not give personal learning help, instant updates or strong exam security, so a better system is needed

5. OBJECTIVE

STUDENT – FOCUSED OBJECTIVE

Develop a centralized LMS platform that allows students to access lecture notes, study materials, and academic resources efficiently.

Monitor and track student performance over time, identifying strengths, weaknesses, and areas requiring improvement.

Provide personalized learning paths and recommendations based on students' performance and engagement in the LMS.

Enhance student engagement and motivation through timely feedback, reminders, and interactive activities such as quizzes and discussion forums.

TEACHER – FOCUSED OBJECTIVE

Automate the evaluation process by calculating grades, aggregating marks, and generating performance reports for individual students.

Ensure academic integrity in online assessments by detecting irregularities and potential cheating during quizzes and tests.

Optimize backend performance and data management using structured databases (e.g., MySQL) for efficient storage, retrieval, and analysis of academic data.

Reduce teacher workload by simplifying administrative tasks, automating performance tracking, and enabling efficient management of large classes.

6. METHODOLOGY

Requirement Analysis:

In this step, the needs of students and teachers are studied.
The system requirements are identified, such as:

- Storing student details and academic records
 - Automatic performance evaluation
 - Providing timely feedback to students
 - Easy access to learning materials
- This step helps in understanding what the system should do.

System Design:

Based on the requirements, the overall system structure is designed.
This includes:

- Designing the database to store marks, users, and course data
 - Defining user roles such as student and teacher
 - Creating simple interfaces for easy navigation
- ER Diagram: [ER Diagram](#)

Development:

In this phase, the actual system is built.

- Frontend is developed using HTML, CSS, and JavaScript for user interaction
 - Backend is developed using PHP or Python to handle logic and data processing
 - MySQL is used to store academic and user data
- Features such as grade tracking, quizzes, notifications, and learning resources are implemented.

Testing:

The system is tested to check whether all features work properly.
Different types of testing are done to:

- Find and fix errors
- Ensure correct evaluation and data storage
- Check system reliability and accuracy

Evaluation and Improvement:

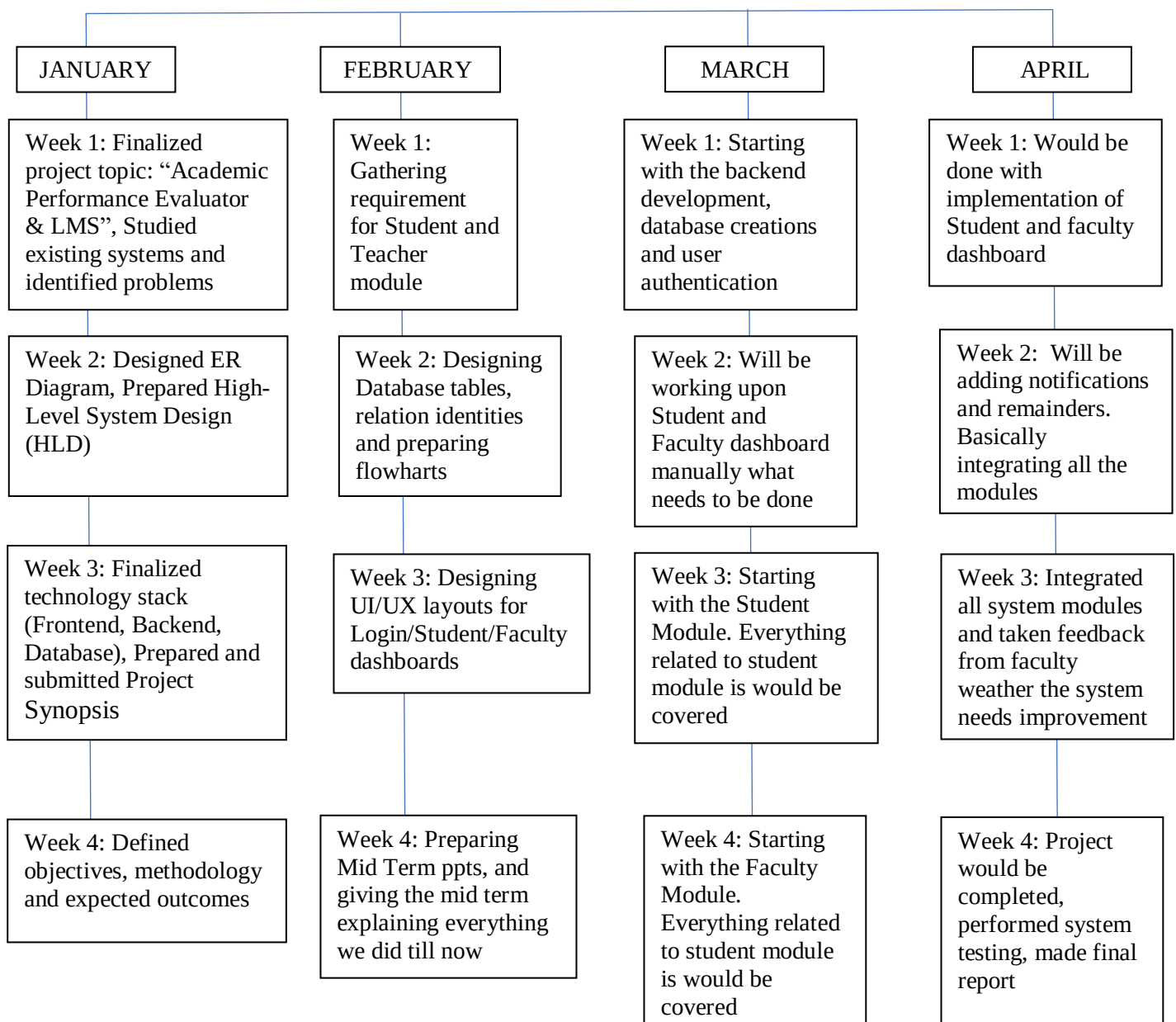
Finally, the system is reviewed based on usability and performance.
Feedback is taken to:

- Improve user experience
- Ensure smooth performance tracking
- Prepare the system for future enhancements

7. EXPECTED OUTCOMES

- Centralized and secure academic data management
- Automated and accurate performance evaluation
- Improved student engagement through timely feedback and reminders
- Reduced manual workload for faculty
- A scalable system adaptable to future enhancements such as analytics

8. PROJECT TIMELINE



BIBLIOGRAPHY

- [1] Pujiati, D., & Averina, M. INTERNATIONAL JOURNAL OF MULTIDISCIPLINARY: APPLIED BUSINESS AND EDUCATION RESEARCH.Center for AI Safety (2023). MI safety course - poisoning attacks reading materials.
- [2] Zhang, X., Obaidat, M. S., Lee, V. C., Xu, D., & Chen, J. (2024, July). An Effective Learning Management System for Revealing Student Performance Attributes. In *2024 International Conference on Computer, Information and Telecommunication Systems (CITS)* (pp. 1-8). IEEE.
- [3] Zhang, X., Obaidat, M. S., Lee, V. C., Xu, D., & Chen, J. (2024, July). An Effective Learning Management System for Revealing Student Performance Attributes. In *2024 International Conference on Computer, Information and Telecommunication Systems (CITS)* (pp. 1-8

